

LandWeb Study-Area Vegetation

SCANFI species composition, and the kNN (v2) to SCANFI (v3) initial-condition change

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⚠ Warning

Draft report: pending human review

Portions of this document were drafted with the help of generative AI (large language models) and have not yet been fully reviewed by the co-authors. Content may be incomplete, inaccurate, or misattributed. This notice will be removed once a complete review has been carried out.

1 Overview

This report summarizes the SCANFI v2 (2020) (Guindon et al. 2026) per-species **crown-closure prevalence** over the **full LandWeb study area** (3,016,183 km², 3226467208 forested 30 m cells). For each SCANFI source species that maps to a simulated LandWeb species it reports the number (and %) of forested study-area cells carrying *any* cover.

One primary purpose is to document the evidence behind the **THUJ_PLI/TSUG_HET → Abie_spp merge** in **LandWeb_preamble**: western redcedar and western hemlock are a *localized* SCANFI over-attribution in the SW Alberta Rockies, not a study-area-wide signal.

It also documents the **change in starting conditions from v2 to v3**: the underlying tree-species product moved from the Beaudoin *k*NN (2001) maps (Beaudoin et al. 2014) to SCANFI (2020) (Guindon et al. 2026) (see [Initial conditions: Beaudoin kNN \(2001\) to SCANFI \(2020\)](#)).

2 Study area composition

Table 1: SCANFI species prevalence over the LandWeb study area, by cells with any cover (sorted). **LandWeb species** is the simulated species each SCANFI code merges into.

SCANFI code	Common name	LandWeb species	Cells (cover > 0)	% of study area
PICE_MAR	black spruce	Pice_mar	1,548,758,984	48.002
LARI_LAR	tamarack	Lari_spp	924,800,678	28.663
POPU_TRE	trembling aspen	Popu_spp	897,899,898	27.829
PINU_BAN	jack pine	Pinu_spp	840,128,756	26.039
PICE_GLA	white spruce	Pice_gla	701,495,401	21.742
BETU_PAP	white birch	Popu_spp	616,449,196	19.106
POPU_BAL	balsam poplar	Popu_spp	290,988,009	9.019
PINU_CON_LAT	lodgepole pine	Pinu_spp	242,557,446	7.518
ABIE_BAL	balsam fir	Abie_spp	152,552,888	4.728
ABIE_LAS	subalpine fir	Abie_spp	89,885,903	2.786
PICE_ENG_GLA	Engelmann x white spruce	Pice_gla	54,901,702	1.702
PICE_ENG	Engelmann spruce	Pice_gla	44,336,103	1.374
PSEU_MEN	Douglas-fir	Pseu_men	10,487,529	0.325

LARI_OCC	western larch	Lari_spp	5,868,751	0.182
PSEU_MEN_GLA	Douglas-fir (interior)	Pseu_men	5,430,224	0.168
THUJ_PLI	western redcedar	Abie_spp	5,248,943	0.163
TSUG_HET	western hemlock	Abie_spp	4,647,291	0.144
PINU_CON_CON	shore pine	Pinu_spp	3,173	0.000

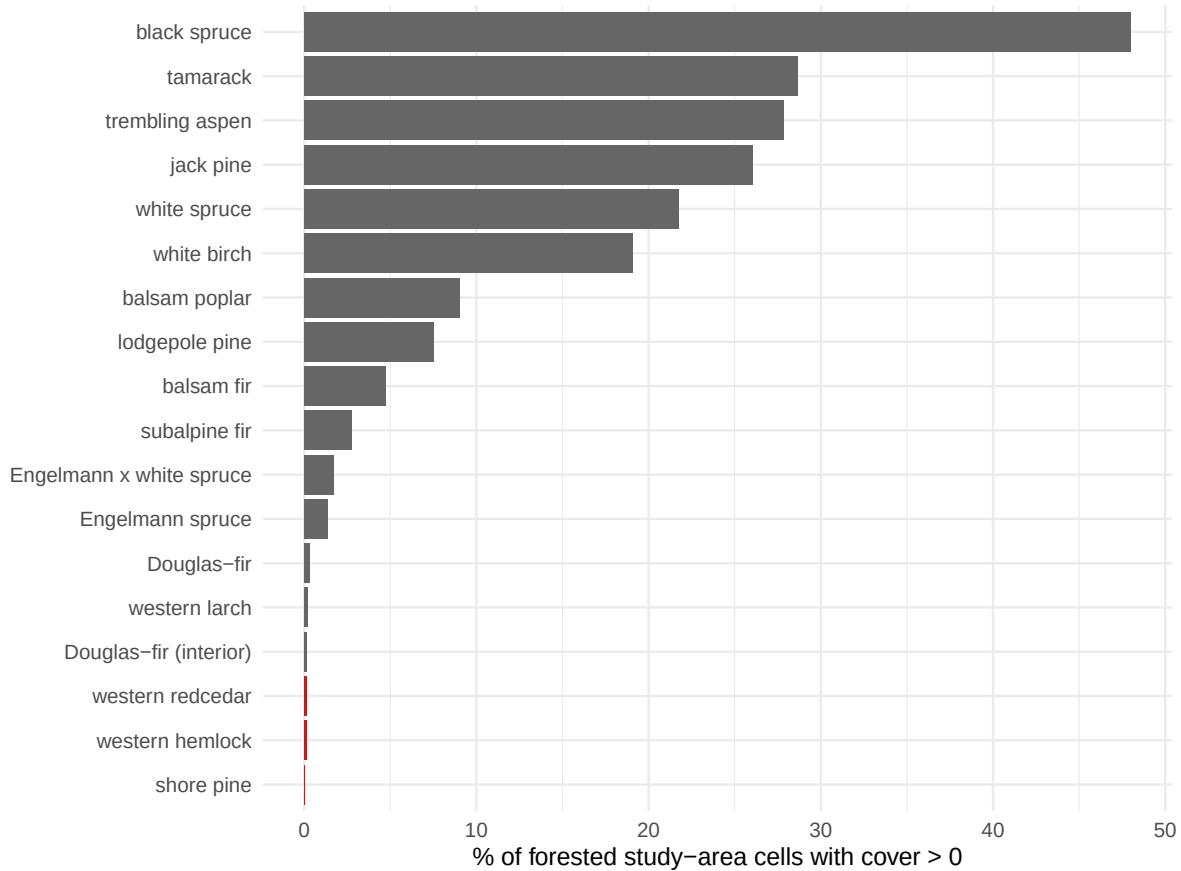


Figure 1: Prevalence (% of forested study-area cells with cover > 0) by SCANFI species; coastal-attribution suspects in red.

The composition is a textbook western-boreal signature — black spruce dominant, then tamarack, aspen, jack pine, and white spruce.

3 Coastal over-attribution (the merge rationale)

Table 2: Coastal-attribution suspects: study-area prevalence and substantial-cover ($\geq 10\%$) extent.

SCANFI code	Common name	% study area (cover > 0)	% study area ($\geq 10\%$)	Max cover %
THUJ_PLI	western redcedar	0.163	0.0373	61
TSUG_HET	western hemlock	0.144	0.0410	70
PINU_CON_CON	shore pine	0.000	0.0000	24

Western hemlock (TSUG_HET) covers only **0.14%** of the LandWeb study area, and western redcedar (THUJ_PLI) **0.16%** — rounding errors at the study-area scale. By contrast, a stand-alone count over the **Spray Lake FMA** (SW Alberta Rockies) put western hemlock on **~25%** of pixels: a $\sim 175\times$ concentration in the SW corner, exactly where SCANFI bleeds coastal species across the BC boundary. Where these species do appear they can reach high cover (up to 70% hemlock), but the footprint is tiny and localized.

Because both fold into **Abie_spp** (the closest shade-tolerant softwood), the merge:

- removes a stand-alone species that has no usable known-age / non-zero-biomass cohorts in the SW — the cause of the **Biomass_borealDataPrep** non-conformable (“code 235”) crash; and
- relabels only $\sim 0.3\%$ of the study area’s cover, leaving the rest untouched.

PINU_CON_CON (shore pine, the coastal *Pinus contorta* variety) is effectively absent across the study area (3,173 cells), so it is left merged into **Pinu_spp**, where — like all pines — it is harmless.

4 Initial conditions: Beaudoin kNN (2001) to SCANFI (2020)

The v2 model built its starting species/biomass/age conditions from the Beaudoin *k*NN maps (Beaudoin et al. 2014) — 250 m, 2001 base year, derived by *k*-nearest-neighbours from MODIS imagery and NFI photo-plots. v3 replaces these with SCANFI (2020) (Guindon et al. 2026; Guindon et al. 2024) at 30 m. Because the model’s initial vegetation is read directly from this product, the switch changes the composition the simulation starts from, in two ways: **which species are resolved**, and **their mapped extent**.

4.1 Species the two products resolve

Table 3: Tree species each inventory resolves, grouped by simulated LandWeb species. SCANFI (v3) resolves larch and Douglas-fir — absent from the kNN product (v2) — and additionally splits Engelmann spruce and the two *Pinus contorta* varieties.

LandWeb species	kNN (v2, 2001)	SCANFI (v3, 2020)
Abie_spp	subalpine fir, balsam fir, fir (unspecified)	balsam fir, subalpine fir, western redcedar, western hemlock
Lari_spp	— (not resolved)	tamarack, western larch
Pice_gla	white spruce	white spruce, Engelmann x white spruce, Engelmann spruce
Pice_mar	black spruce	black spruce
Pinu_spp	jack pine, lodgepole pine	jack pine, lodgepole pine, shore pine
Popu_spp	trembling aspen, white birch	trembling aspen, white birch, balsam poplar
Pseu_men	— (not resolved)	Douglas-fir, Douglas-fir (interior)

The kNN product resolves **5** LandWeb species groups; SCANFI resolves **7**. The two SCANFI adds are consequential: **Lari_spp** and **Pseu_men** — i.e. **larch/tamarack** and **Douglas-fir**, which have no kNN layer at all. SCANFI also resolves Engelmann spruce (folded into **Pice_gla**) and both *Pinus contorta* varieties, where kNN carried a single lumped fir group and a single lodgepole layer.

4.2 Composition shift

Table 4: Beaudoin kNN (2001) per-species mean cover and prevalence over the LandWeb study area (sorted by mean cover).

kNN species	LandWeb species	Mean cover %	% of domain (cover > 0)
black spruce	Pice_mar	26.67	62.84
trembling aspen	Popu_spp	16.41	53.60
jack pine	Pinu_spp	8.03	38.28

white spruce	Pice_gla	4.58	36.33
lodgepole pine	Pinu_spp	2.24	10.37
white birch	Popu_spp	1.63	27.70
subalpine fir	Abie_spp	1.15	4.30
balsam fir	Abie_spp	0.37	8.32
fir (unspecified)	Abie_spp	0.11	0.66

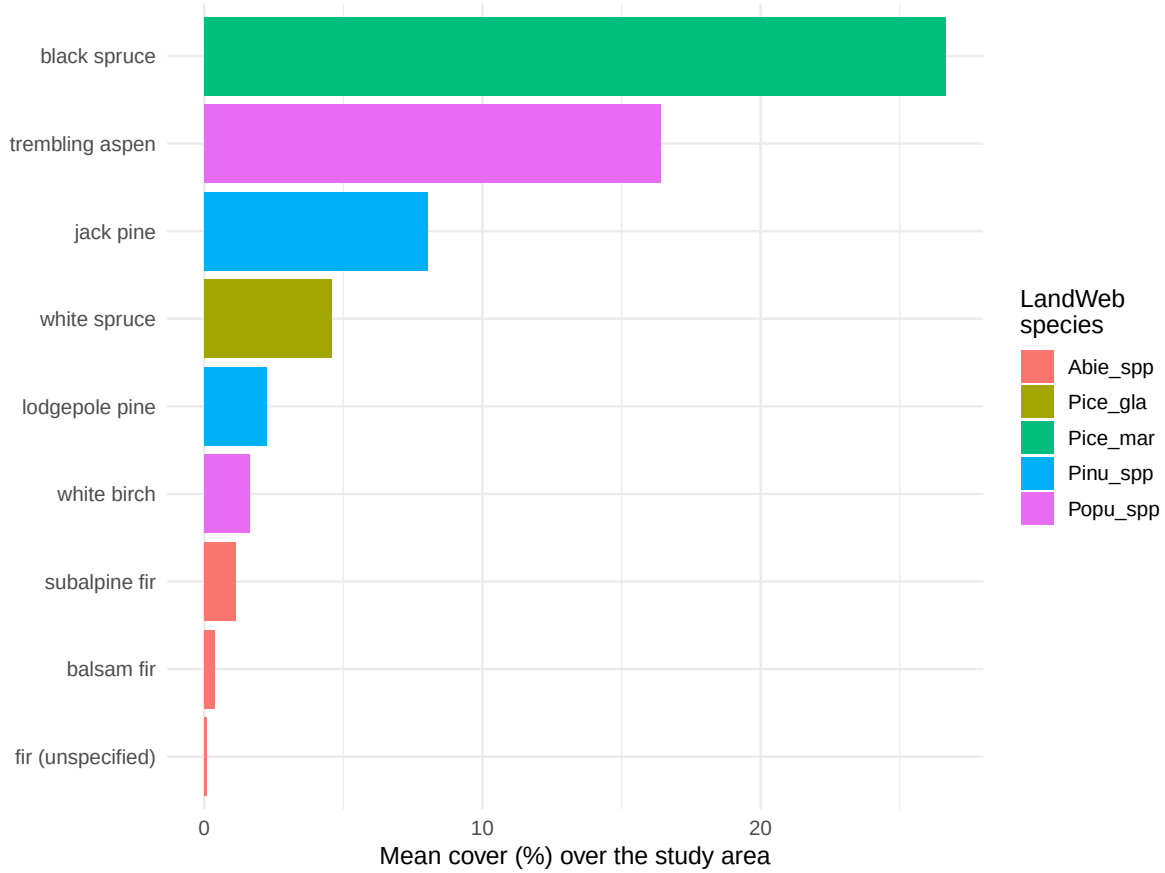


Figure 2: Beaudoin kNN (2001) mean cover by species over the LandWeb study area, coloured by simulated LandWeb species. The kNN product carries no larch or Douglas-fir layer.

Both products agree on the dominant signature — **black spruce** leads by a wide margin, followed by aspen, jack pine, and white spruce. The most consequential difference is at #2: in SCANFI, **tamarack/larch (Lari_spp)** is the **second most prevalent group** (on **28.8%** of the study area), yet the kNN product has **no larch layer** — so the v2 model started with that entire LandWeb species group absent. The two products' resolutions (250 m vs 30 m) make the raw prevalence values not directly comparable in magnitude, so the comparison here emphasizes the species set and relative ordering rather than absolute percentages.

4.3 Bearing on simulated outcomes

Because the initial cohort composition is seeded from this product, the v3 starting landscape carries a **richer species pool** (7 vs 5 LandWeb groups) with larch and Douglas-fir newly present, Engelmann spruce distinguished, and a two-decade-newer vintage (2001 to 2020) that already reflects the intervening disturbance and growth. These initial-condition differences are one of the contributors to any change in simulated outcomes between v2 and v3, alongside the fire-input update (report 02) and the code/module changes (report 03); the output comparison (report 04) is where their combined effect is quantified.

i Note

Scope of this comparison. This section compares *species composition* only. Two related initial-condition changes are out of scope here: (1) the **stand-age and biomass** layers (kNN 250 m vs SCANFI 30 m), whose study-area-wide comparison needs the gated 30 m scan; and (2) the **current-condition** update from the earlier CASFRI/Pickell-era layers to the 2026 merged Alberta AVI + BC VRI + national SBFI composite, which is pending that composite being staged locally.

References

- Beaudoin, A, P Y Bernier, L Guindon, et al. 2014. “Mapping Attributes of Canada’s Forests at Moderate Resolution Through kNN and MODIS Imagery.” *Canadian Journal of Forest Research* 44: 521–32. <https://doi.org/10.1139/cjfr-2013-0401>.
- Guindon, L., D. L. P. Correia, F. Manka, and B. Smiley. 2026. *SCANFI V2: Spatialized CANadian National Forest Inventory Data Product V2*. Natural Resources Canada, Canadian Forest Service, Laurentian Forestry Centre, Quebec, Canada. <https://doi.org/10.23687/07653869-f303-46c2-a04e-9ab479b73cbf>.
- Guindon, Luc, Francis Manka, David L. P. Correia, et al. 2024. “A New Approach for Spatializing the Canadian National Forest Inventory (SCANFI) Using Landsat Dense Time Series.” *Canadian Journal of Forest Research* 54 (7): 793–815. <https://doi.org/10.1139/cjfr-2023-0118>.

Provenance

- **Sources:** SCANFI v2 species crown-closure layers (Guindon et al. 2026), derived via the SCANFI approach (Guindon et al. 2024) (`SCANFI_spsCC_<code>_2020_v2_20260119.tif`); and the Beaudoin *k*NN 2001 species layers (Beaudoin et al. 2014) (`NFI_MODIS250m_2001_kNN_Species_<Sp>`). Both masked to the LandWeb study area (the LTHFC v10 union, `landweb_lthfc_v10`).
- **Data files:** `outputs/extended_analyses/scanfi_lthfc_v10_2020_veg_summary.csv` (SCANFI); `outputs/extended_analyses/knn_lthfc_v10_2001_veg_summary.csv` (kNN).
- **Method:** `R/scanfi_veg_summary.R` (multi-hour over the ~ 3.0 M km² study area at 30 m) and `R/knn_veg_summary.R` (~ 40 s at 250 m), both cache-aware `format = "file"` targets built by the gated `landweb.extended_analyses` factory; reused unless the CSV is deleted or a version stamp in its name changes.

Appendix: Build provenance

This appendix pins the rendered PDF to the code, environment, and revisions that produced it, so a result can be reproduced or a difference attributed to code vs. inputs vs. randomness. It is generated at render time from the `workflowtools info_*`() family.

Build identity. LandWeb repository, branch `targets`, commit `ad683998`, rendered 2026-07-03 10:15 MDT with R version 4.6.1 (2026-06-24) on Ubuntu 26.04 LTS (GEOS 3.14.1, GDAL 3.12.2, PROJ 9.7.1).

Table 5: SpaDES modules and owned packages at their pipeline-pinned revisions when this report was rendered (`workflowtools::info_submodules()`).

Component	Kind	Pinned SHA
Biomass_borealDataPrep	module	f570094ea9
Biomass_core	module	32a3790870
Biomass_regeneration	module	b739da2d32
Biomass_speciesData	module	101ad29571
Biomass_speciesFactorial	module	01d4559d15
Biomass_speciesParameters	module	1e381a7881
HSI_Caribou_MB	module	1180d61743
LandMine	module	d7ccb083e1
LandWeb_output	module	af78cee648
LandWeb_preamble	module	65dc365f37
LandWeb_summary	module	99e229e3ac
NRV_summary	module	ca7b444fa2
burnSummaries	module	add6cc0bb4
timeSinceFire	module	30d242ea7b
LandWebUtils	package	cf29ea06e4
SpaDES.targets	package	b1f68a5467
crew.ssh	package	79c368f0fa
nrvtools	package	f9665fd7b0

spatialutils	package	a891aef9d5
workflowtools	package	561def4389

The full `sessionInfo()` and the exact `renv.lock` package pins accompany the run; input datasets are recorded in `_input_manifest.json` (report 06).

DRAFT